Data Science Project Report :

Students (Team) :

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Introduction:

This project uses a retail transactions dataset. The dataset includes columns such as Transaction ID, Item, Quantity, Price per Unit, Total Spent, Payment Method, Location, and Transaction Date. The goal is to clean the dataset, perform exploratory data analysis (EDA), and build a regression AI module to predict TotalSpent.

Questions about the dataset:

- Q1) What are the top 10 most frequently sold items?
- Q2) What is the distribution of Payment Method and Location?
- Q3) What are the distributions of Quantity, Price_Per_Unit, and TotalSpent?
- Q4) What is the relationship between Quantity and TotalSpent, and what are the correlations betweennumeric variables?
- Q5) Can we predict TotalSpent using a refined Linear Regression AI module?

Import Libraries:

```
import numpy as np import pandas as pd import matplotlib.pyplot as plt import seaborn
as sns
```

Data Wrangling / Data Cleaning:

Dataset preview before cleaning (first 10 rows):

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
TXN_4717867		5	3.0	15.0		Takeaway	2023-07-28
TXN_2064365	Sandwich	5	4.0	20.0		In-store	2023-12-31

Shape before cleaning: (10000, 8)

Missing values (top 8) before cleaning:

Column	Missing
Location	3265
Payment Method	2579
Item	333
Price_Per_Unit	179
TotalSpent	173

Transaction Date	159
Quantity	138
Transaction ID	0

Cleaning steps applied (based on the provided code):

• Normalize all text columns (lowercase + strip).

• Replace bad tokens (unknown, error, '-', etc.) with NaN.

• Drop missing Transaction ID and remove duplicate Transaction IDs.

• Clean Item and impute missing values with 'unknown_item'.

• Convert numeric columns and handle outliers using IQR (Quantity, Price_Per_Unit).

• Impute Quantity and Price_Per_Unit using median.

• Smart-impute TotalSpent using Quantity × Price_Per_Unit, then median.

• Smart-impute Payment Method using the observed distribution (fixed random seed).

• Impute Location using mode.

• Parse Transaction Date and impute missing dates with mode.

• Remove full-row duplicates.

Key cleaning outputs:

• Missing Transaction ID before dropna: 0

• Missing Transaction ID after dropna: 0

• Duplicate Transaction ID after de-duplication: 0

- Item missing after cleaning: 969
- IQR outliers set to NaN: Quantity=0, Price_Per_Unit=0
- Missing Quantity after cleaning: 479 (final missing: 0)
- Missing Price_Per_Unit after cleaning: 533 (final missing: 0)
- Missing TotalSpent before smart impute: 502 (final missing: 0)
- Final missing Payment Method: 0
- Final missing Location: 0
- Missing Transaction Date after parsing: 460 (final missing: 0)
- Removed full-row duplicates: 0

Dataset after cleaning:

Dataset preview after cleaning (first 10 rows):

Transaction ID	Item	Quantity	Price_Per_Unit	TotalSpent	Payment Method	Location	Transaction Date
txn_1961373	coffee	2.0	2.0	4.0	credit card	takeaway	2023-09-08
txn_4977031	cake	4.0	3.0	12.0	cash	in-store	2023-05-16
txn_4271903	cookie	4.0	1.0	4.0	credit card	in-store	2023-07-19
txn_7034554	salad	2.0	5.0	10.0	credit card	takeaway	2023-04-27
txn_3160411	coffee	2.0	2.0	4.0	digital wallet	in-store	2023-06-11
txn_2602893	smoothie	5.0	4.0	20.0	credit card	takeaway	2023-03-31
txn_4433211	unknown_item	3.0	3.0	9.0	cash	takeaway	2023-10-06
txn_6699534	sandwich	4.0	4.0	16.0	cash	takeaway	2023-10-28
txn_4717867	unknown_item	5.0	3.0	15.0	cash	takeaway	2023-07-28
txn_2064365	sandwich	5.0	4.0	20.0	credit card	in-store	2023-12-31

Shape after cleaning: (9672, 8) **Missing values (top 8) after cleaning:**

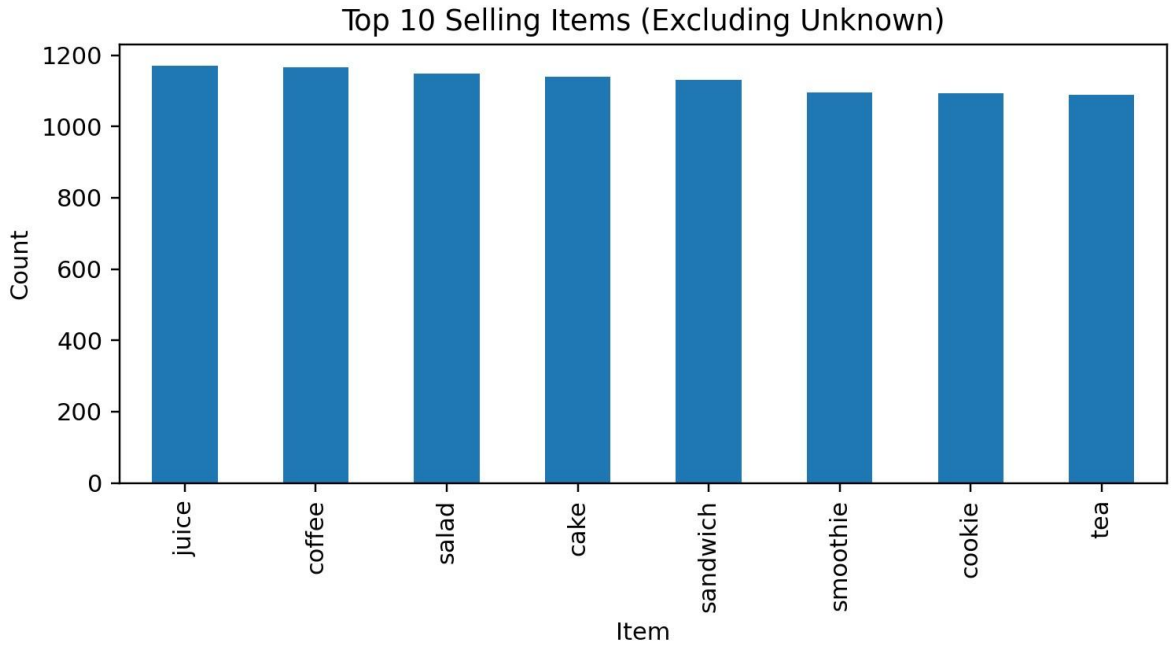
Column	Missing
Transaction ID	0
Item	0
Quantity	0
Price_Per_Unit	0
TotalSpent	0
Payment Method	0

Location	0
Transaction Date	0

Exploration with Visuals and Conclusions:

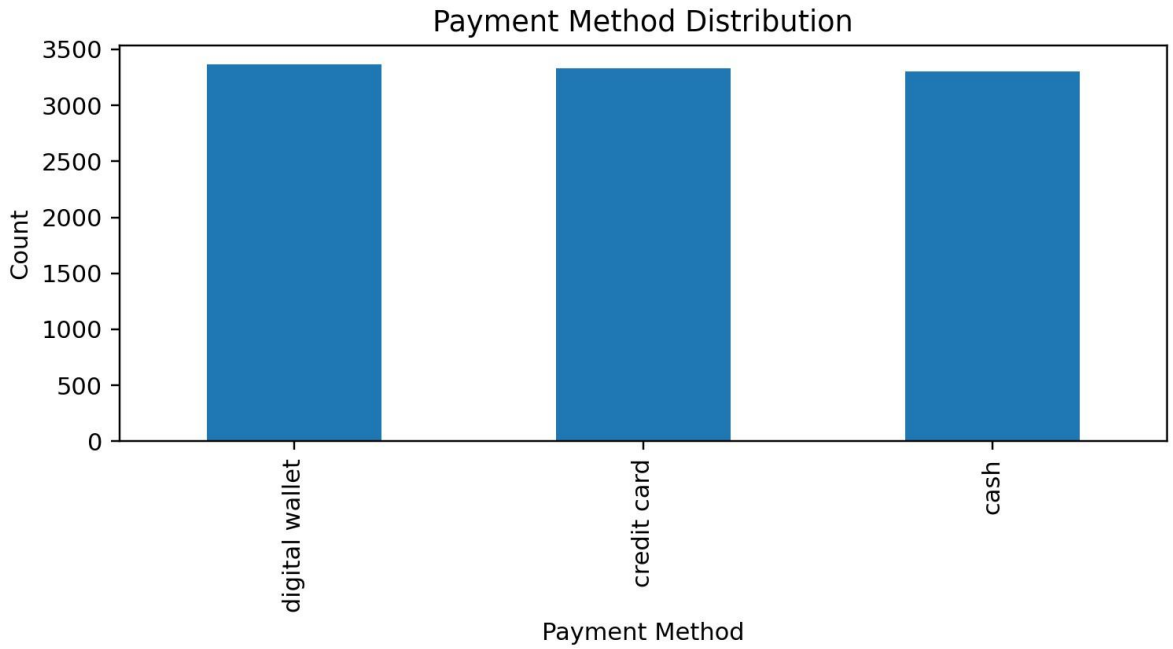
We answer the questions using EDA plots and short conclusions (same style as the example document).

Q1: Top 10 Selling Items:



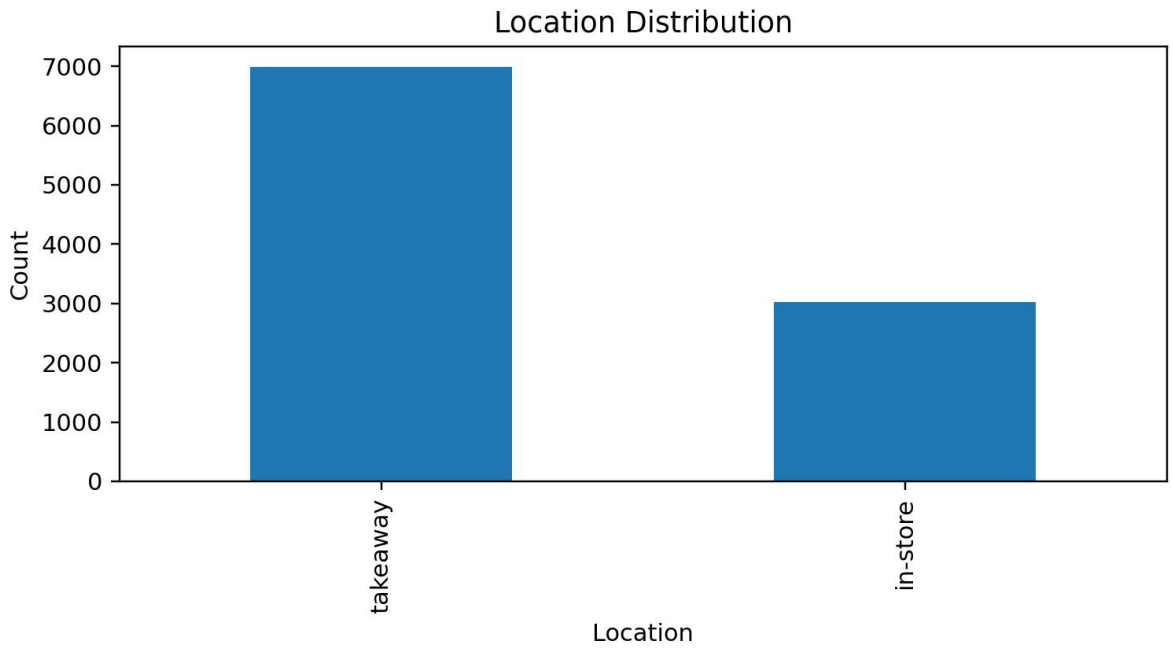
Conclusion: This bar chart shows the 10 most frequently sold items. It helps identify best-sellers and inventory focus.

Q2: Payment Method Distribution:



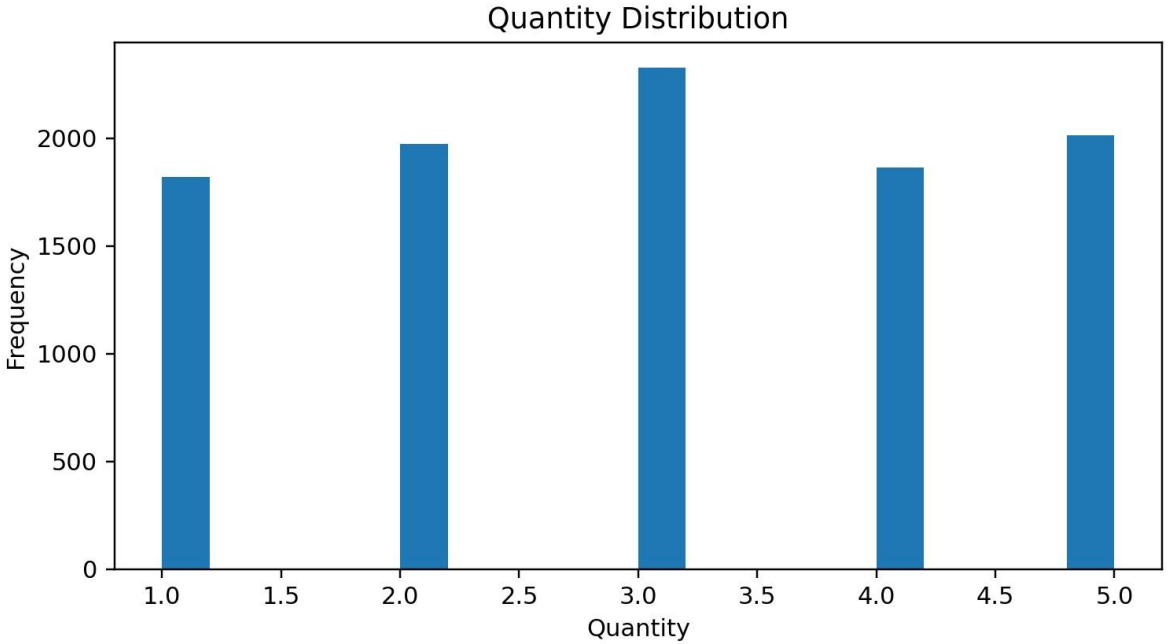
Conclusion: This chart describes how customers prefer to pay. The dominant bar indicates the most common payment method.

Q2: Location Distribution:



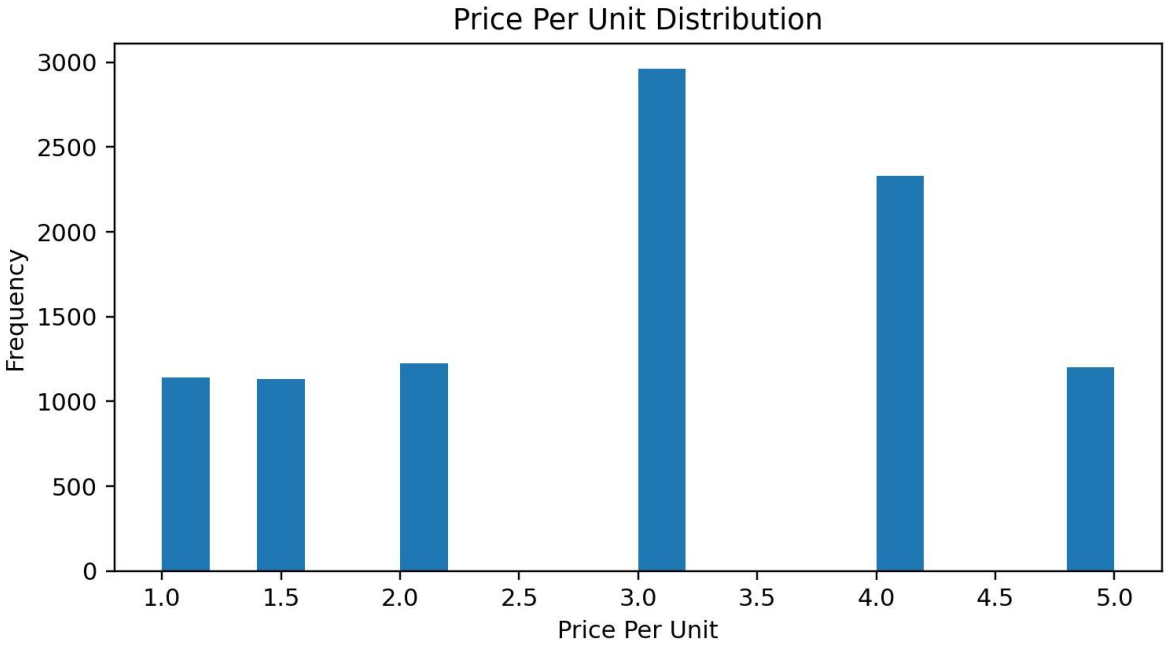
Conclusion: This plot shows transactions by location. It gives insight into where purchases happen most.

Q3: Quantity Distribution :



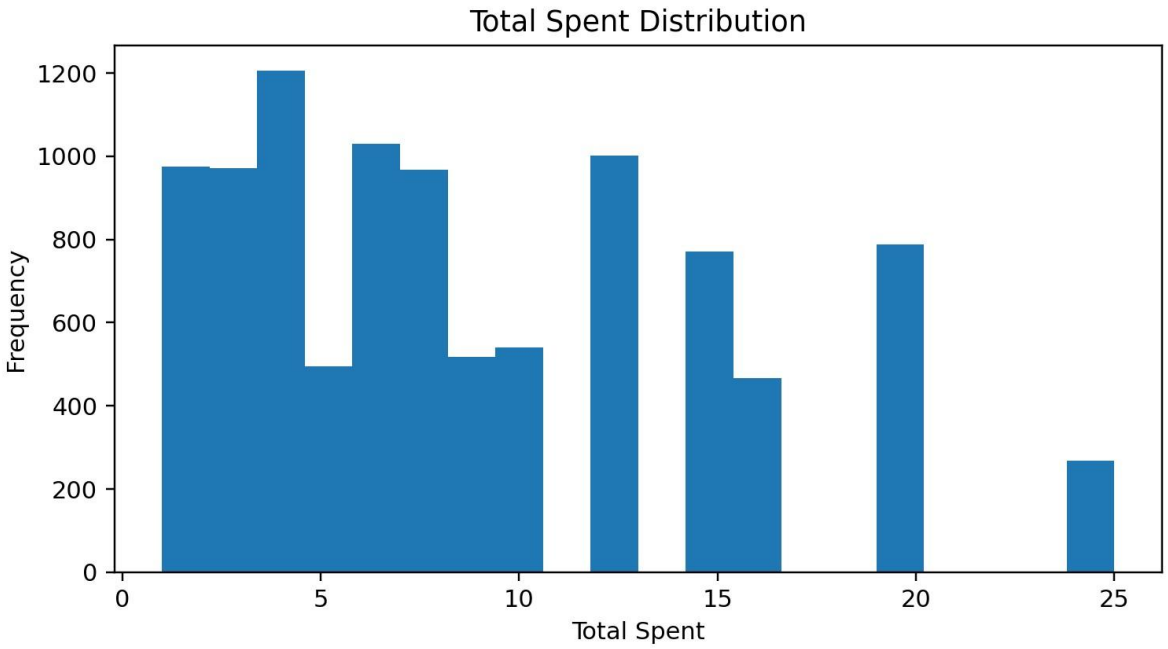
Conclusion: The histogram shows how quantities are distributed. Higher bars at small values indicate typical low-quantity purchases.

Q3: Price Per Unit Distribution:



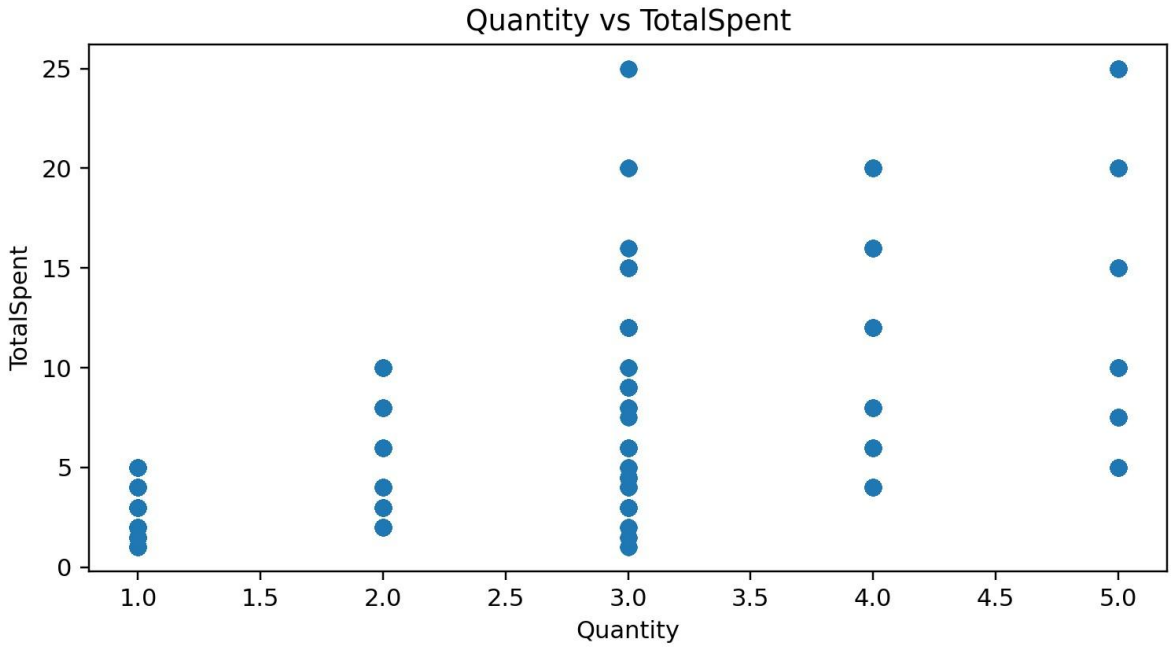
Conclusion: This histogram shows the price range of items. Peaks indicate common price points.

Q3: Total Spent Distribution:



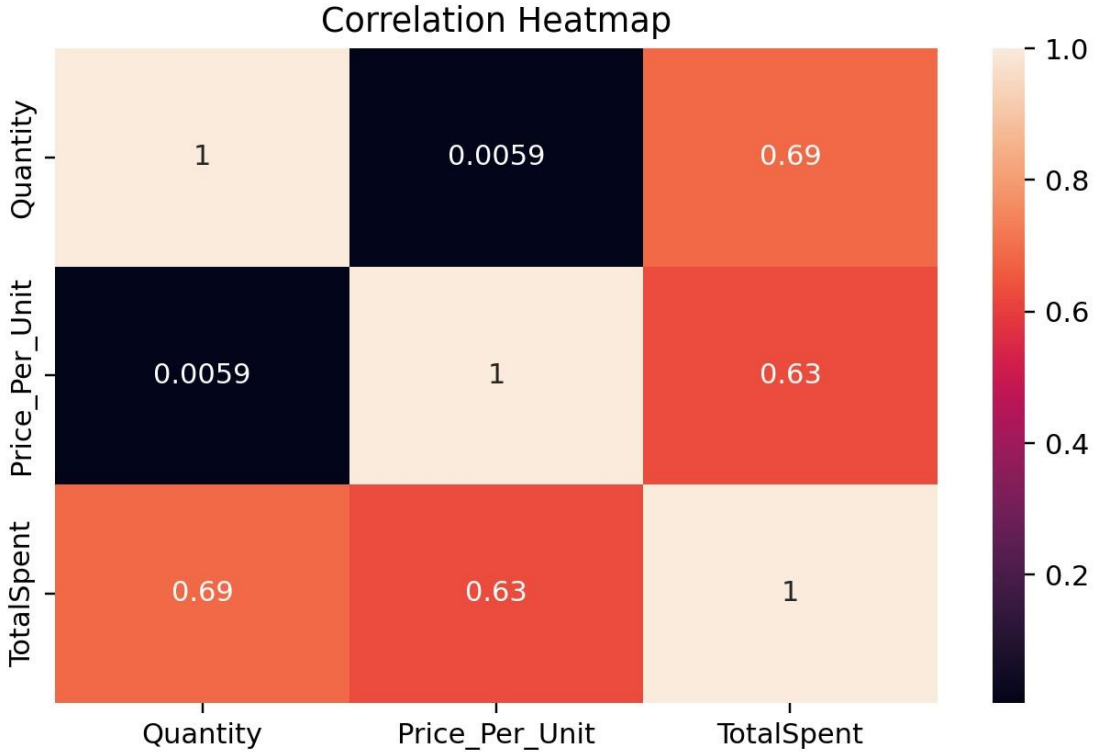
Conclusion: TotalSpent usually has a wider spread since it depends on both quantity and unit price.

Q4: Quantity vs TotalSpent:



Conclusion: The scatter plot shows a positive relationship: as Quantity increases, TotalSpent tends to increase.

Q4: Correlation Heatmap:



Conclusion: This heatmap summarizes linear correlations. TotalSpent should correlate with Quantity and Price_Per_Unit.

AI Module Used:

Type: **Regression** (because TotalSpent is continuous). AI Module: **Linear Regression (Refined)** with StandardScaler for numeric features, OneHotEncoder for Payment Method, and engineered features (day, month, weekday, and QxP = Quantity × Price_Per_Unit).

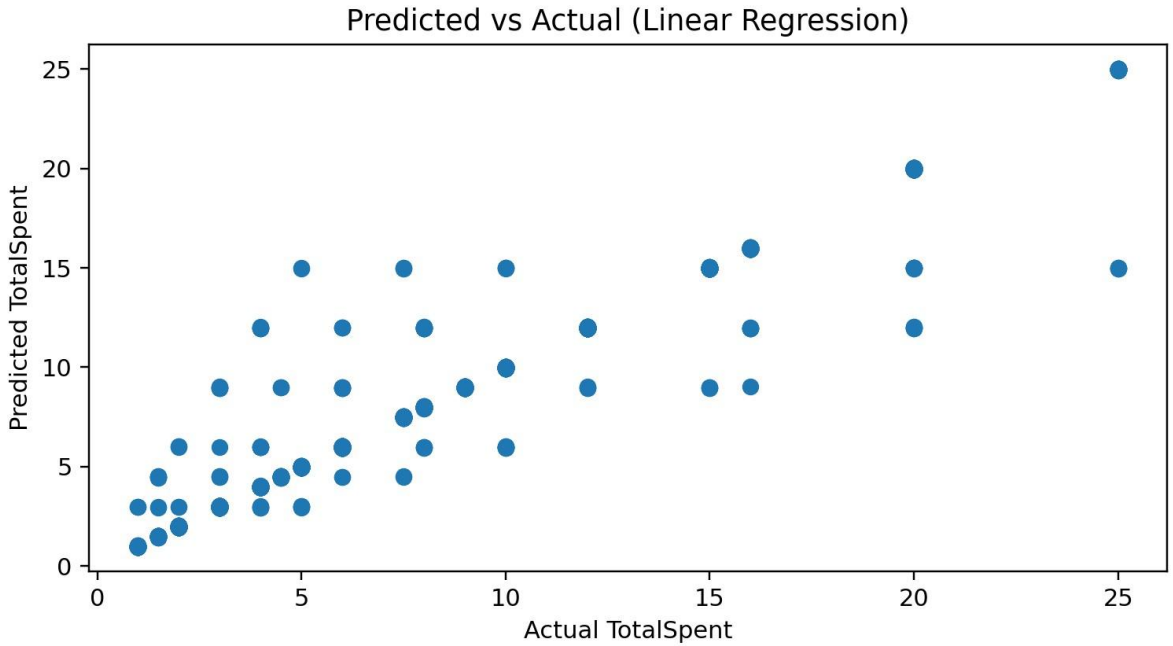
Metric	Value
MAE	0.578
RMSE	1.8
R2	0.90

Conclusion: Higher R2 is better; lower MAE/RMSE means smaller prediction errors.

Model Outputs (Plots):

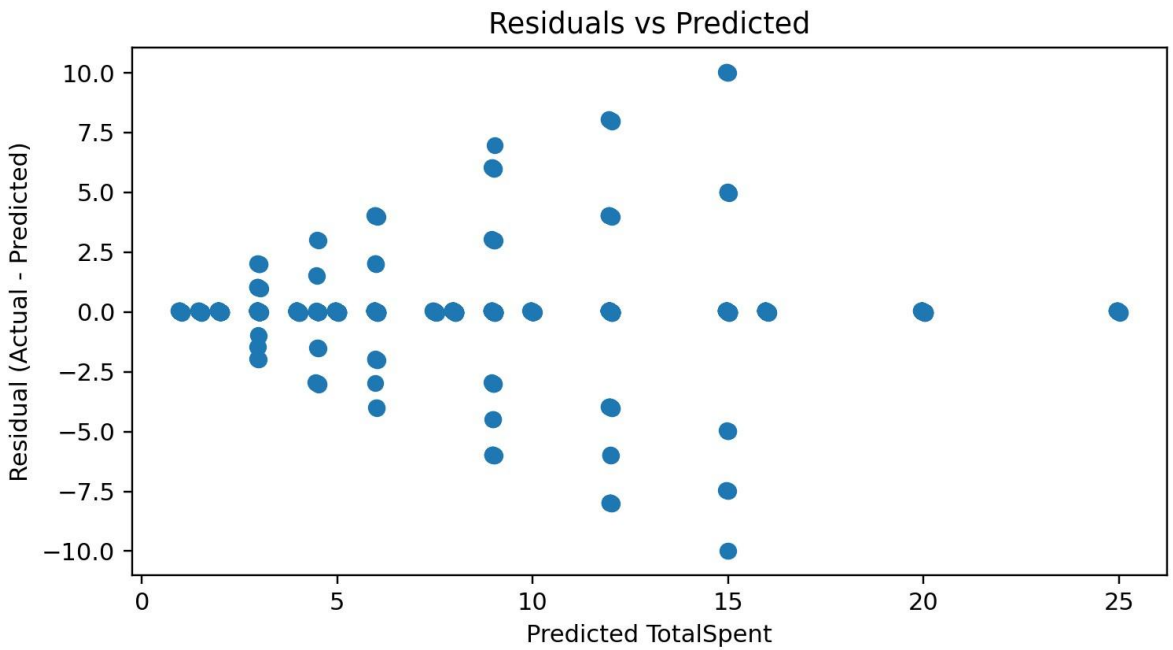
Visual evaluation of the Linear Regression model.

Model Plot: Predicted vs Actual :



Conclusion: Points closer to the diagonal indicate better predictions; wide spread indicates larger error.

Model Plot: Residuals:



Conclusion: Residuals should be randomly scattered around zero. Patterns may indicate missing non-linear effects.

Final Conclusion:

The dataset was cleaned by handling invalid tokens, missing values, outliers (IQR), and duplicates. EDA revealed best-selling items, customer payment preferences, and a positive relationship between Quantity and TotalSpent. Finally, a refined Linear Regression regression module was trained to predict TotalSpent.

Limitations:

Results depend on this dataset and one train/test split. Linear Regression may underfit if relationships are non-linear. Cross-validation and additional features could improve generalization.

References:

• Pandas Documentation
• Scikit-learn Documentation
• Matplotlib & Seaborn Documentation